MIL-P-46575B (MU)
15 November 1963
SUPERSEDING
MIL-P-46575A (MU)
30 January 1963

MILITARY SPECIFICATION

PRIMER, PERCUSSION-ELECTRIC, M67 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of primer, designated as Primer, Percussion-Electric, M67, for use in Artillery Ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223 - Powder, Black
MIL-A-2550 - Ammunition and Special Weapons, General Specification for.
MIL-I-45208 - Inspection Requirement; General Specification for.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.
MIL-I-46574 - Igniter, Percussion-Electric, M30, Loading, Assembling and Packing.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.

FSC: 1390

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:21Z
Check the source to verify that this is the current version before use.

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

7548520 - Primer, Percussion, Electric M67, Loading Assembly
8798866 - Box, Packing, Ammunition, for Primer, Percussion, Electric M67, or Primer, Percussion, Electric, T87E1.
8798867 - Carton, Packing, Ammunition for Primer, Percussion, Electric M67, or Primer, Percussion, Electric, T87E1.

PUBLICATIONS

ORDNANCE CORPS

IEL-7548520 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Primers.-The primers shall comply with all requirements specified on Drawing (dwg.) 7548520 and with all requirements specified in applicable specifications.

3.3 Moisture content of black powder.-The moisture content of the black powder at the loading station, at the time of loading, shall not exceed 0.5 percent when determined as specified in 4.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:21Z
Check the source to verify that this is the current version before use.

3.4 Electrical resistance.-The primers shall have an electrical resistance of not less than 0.80 ohms nor more than 1.80 ohms when tested as specified in 4.3.2.

* 3.5 Functioning (Electrical and percussion).-The primer shall comply with the following requirements when tested as specified in 4.3.3.

a. The primer shall not fail to fire.
b. The primer body shall not rupture or become detached.
c. There shall be no audible hangfire. (See 6.4)
d. The primer shall not squib. (See 6.4)

3.6 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. All packing components shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:21Z
Check the source to verify that this is the current version before use.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analysis on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:21Z
Check the source to verify that this is the current version before use.

4.2 Inspection provisions

4.2.1 Lot formation.-A lot shall consist of loaded primers produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot number. Each lot shall contain:

a. Percussion electric igniter assemblies of one lot interfix number from one manufacturer.

b. Primer metal components of one lot interfix number from one manufacturer.

c. Black powder of one lot interfix number from one manufacturer.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Assembly, prior to loading (see dwg. 7548469 covering a detail of dwg. 7548520).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:			
1.	Any flash hole in body missing....	Gage	01001
Major:	AQL 0.25 percent		
101.	Component damaged.....	Visual	01002
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	01003

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:21Z
Check the source to verify that this is the current version before use.

4.2.2.2 Assembly, prior to inserting diaphragm (see dwg. 7548520).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Weight of charge	Balance	02001
102.	Any flash hole not completely sealed (see Note 1)	Visual	02002
103.	Liner fails to cover all holes ..	Visual	02003
Minor:	AQL 0.40 percent		
201.	Foreign matter in charge	Visual	02004

Note 1 - Sketches 1, 2 and 3 on Figure 1 indicate acceptable primers. Furthermore, a primer with holes displaying the conditions 4, 5 or 6 may be classed acceptable provided that the visible liner is thoroughly impregnated with paint or lacquer and not more than 20 percent of the holes have this condition.

4.2.2.3 Assembly (see dwg. 7548520).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Depth to diaphragm	Gage	03001
102	Igniter above flush	Straight-edge	03002
103.	Stake missing	Visual	03003
104.	Diaphragm or liner missing, punctured or improperly sealed ..	Visual	03004
105.	Charge missing	Manual	03005
106.	Split or cracked head	Visual	03006
107.	Diaphragm inverted	Visual	03007

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:21Z
Check the source to verify that this is the current version before use.

Minor:	AQL 1.50 percent		
201.	Total length, maximum (max.)	Gage	03008
202.	Marking misleading or unidentifiable	Visual	03009
203.	Evidence of poor workmanship (see 3.6)	Visual	03010

4.2.2.4 Carton, prior to sealing (see dwg. 8798867).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Any packing material missing	Visual	04001
102.	Separator damaged to the extent that primers touch each other ...	Visual	04002
103.	Primer missing	Visual	04003
Minor:	None defined		

4.2.2.5 Carton, sealed (see dwg. 8798867).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.50 percent		
201.	Sealing strip torn, badly wrinkled, incomplete or otherwise fails to seal carton completely	Visual	05001
202.	Carton torn, cut or punctured ...	Visual	05002
203.	Contents loose	Manual	05003
204.	Marking missing, misleading or unidentifiable	Visual	05004

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:21Z
Check the source to verify that this is the current version before use.

4.2.2.6 Box, packing prior to sealing (see dwg. 8798866).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Carton missing	Visual	06001
102.	Bag missing (when required)	Visual	06002
103.	Bag punctured, torn or improperly sealed (when required)	Visual	06003
Minor:	None defined		

4.2.2.7 Box, packing, sealed (see dwg. 879866).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Box, broken, board loose, or protruding nail	Visual	07001
102.	Strapping missing, broken or loose	Visual-Manual	07002
103.	DOD symbol missing or unidentifiable	Visual	07003
Minor:	AQL 1.50 percent		
201.	Contents loose	Manual	07004
202.	Strapping improperly engaged or mislocated	Visual	07005
203.	Marking misleading or unidentifiable	Visual	07006

4.2.3 Testing

4.2.3.1 Moisture content of black powder (see 3.3) - Major defect - Code No. 08001.-The contractor shall provide adequate controls to assure the compliance of each lot with the applicable

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:21Z
Check the source to verify that this is the current version before use.

requirement. For verification, and just prior to start of each eight hours loading production, the contractor shall test one sample of powder selected from that quantity, or sub-lot of powder that will be used in the eight hour loading production of primer assemblies. Composite samples shall not be used. If the moisture content of the sample exceeds the requirement, and loading has not begun, that sub-lot of powder represented by the sample shall be rejected. If powder from a sub-lot with excessive moisture has been used in loading the remaining unloaded powder and all assemblies containing the non-conforming powder shall be rejected.

4.2.3.2 Electrical resistance (see 3.4) - Major defect - Code No. 03011.-This test shall be conducted 100 percent. Any primer which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.3.2 using equipment in accordance with 4.2.4.

4.2.3.3 Functioning (Proving ground) - (See Table I)

TABLE I

Defect	Classification	Code No.
Primer Fails to Fire (see 3.5)	Major	03012
Primer Body Ruptures or Becomes Detached (see 3.5)	Critical	03013
Audible Hangfire (see 3.5)	Major	03014
Primer Squibs (see 3.5)	Major	03015

4.2.3.3.1 Percussion phase (see 3.5 and Table I)

4.2.3.3.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, the acceptance and rejection criteria for the test shall be in accordance with Standard MIL-STD-105, Single sampling plan, Code letter H, and an AQL of 1.50 percent. The lot shall be rejected, if a critical defect occurs, or the number

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:21Z
Check the source to verify that this is the current version before use.

of assemblies that exhibit major defects equals or exceeds the sampling plan rejection number (see Table I for defect classification). The test shall be performed as specified in 4.3.3.1.

4.2.3.3.1.2 After three consecutive lots have met the criteria of 4.2.3.3.1.1, the acceptance and rejection criteria shall be in accordance with Standard MIL-STD-105, Single sampling plan, Code letter E, and an AQL of 1.50 percent.

4.2.3.3.2 Electrical phase (see 3.5 and Table I).

4.2.3.3.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, the acceptance and rejection criteria for this test shall be in accordance with Standard MIL-STD-105, Single sampling plan, Code letter H, and an AQL of 1.50 percent. The lot shall be rejected, if a critical defect occurs, or the number of assemblies that exhibit major defects equals or exceeds the sampling plan rejection number (see Table 1 for defect classification). The test shall be performed as specified in 4.3.3.2.

4.2.3.3.2.2 After three consecutive lots have met the criteria of 4.2.3.3.2.1, the acceptance and rejection criteria shall be in accordance with Standard MIL-STD-105, Single sampling plan, Code letter E, and an AQL of 1.50 percent.

4.2.3.4 Check test for deterioration of the M30 percussion electric igniters.-If the total elapsed time between the original acceptance of any percussion electric igniter lot and the assembly of that lot into the primers exceeds two years, or if the igniters have been exposed to adverse conditions, however brief, at any time since previous tests, the percussion electric igniterrr lot shall be subjected to and must satisfactorily pass the check test specified in the applicable percussion electric igniter loading specification, immediately before the igniters are assembled into the primers. The test shall be performed by the contractor on percussion electric igniters selected by the Government inspector at the facility assembling the igniters into the primers (see 6.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:21Z
Check the source to verify that this is the current version before use.

4.2.4 Inspection Equipment.-Index of Inspection Equipment Number IEL 7548520 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 Determination of moisture content of black powder.-The moisture content of the black powder shall be determined in accordance with the procedure specified in Specification MIL-P-223.

4.3.2 Electrical resistance.-The primers shall be tested for electrical resistance by applying an electrical circuit which limits the current through the primer to not more than 20 milliamperes with the igniter plunger as one contact and the primer head as the other.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:21Z
Check the source to verify that this is the current version before use.

4.3.3 Functioning (Proving ground)

4.3.3.1 Percussion.-This test shall be performed at a Government proving ground. The sample primers shall be assembled to complete rounds for which the primer was designed and percussion fired from a gun for which the round is standard. Observation shall be made for compliance with the applicable requirements.

4.3.3.2 Electrical.-This test shall be performed as specified in 4.3.3.1 except that the primers shall be electrically fired. Observation shall be made for compliance with the applicable requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level B.-The primer shall be preserved and packaged (with bag) in accordance with dwg. 8798867.

5.1.2 Level C.-The primer shall be preserved and packaged (without bag) in accordance with dwg. 8798867).

5.2 Packing

5.2.1 Level B.-The cartons shall be packed in accordance with dwg. 8798866.

5.3 Marking.-Marking shall be in accordance with dwgs. 8798867 and 8798866.

5.4 Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:21Z
Check the source to verify that this is the current version before use.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 Inspection Equipment Lists.-Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection Equipment List codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the Inspection Equipment List by the following codes:

CDOF - Contractor's design responsibility on Ordnance format; ORDM-608-12.
CDCF - Contractor's design responsibility on contractor format.
OD - Ordnance design.
ODMU - Ordnance design, mandatory for use.
CE - Commercial equipment.
SCD - Specification Control Drawing.

6.3.3 Ordnance designs.-Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and non-mandatory. When an Inspection Equipment List references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:21Z
Check the source to verify that this is the current version before use.

design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawings.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:21Z
Check the source to verify that this is the current version before use.

CDOF shall comply with the format and requirements of ORDM 608-12, and, in addition, contain the detail and information specified above.

6.3.7 Submission of contractor designs.-Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Definitions:

a. Hangfire.-A failure wherein an audible delay occurs between the instant of application of energy and the detonation report.

b. Squib.-A failure wherein a burning of the primer mixture occurs without detonation.

6.5 Cost of check test.-The contracting officer shall arrange to have the contractor reimbursed for any expenses incurred in the performance of the check test for possible deterioration.

6.6 Check test for possible deterioration of percussion electric primers.-The following tests are designed to prevent the loading of major ammunition items with percussion electric primers which have deteriorated. The tests shall be performed on each lot of percussion electric primers which has been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the percussion electric primer is assembled does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check test in the contract.

6.6.1 Moisture content of black powder.-A sample of 25 primers shall be selected for this test. The moisture content of the black powder shall not exceed 0.7 percent at the time of the performance of the check test for deterioration. If one or more of the sample primers contain black powder wherein the moisture content exceeds 0.7 percent the lot shall be rejected. The test shall be performed as specified in Specification MIL-P-223.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:21Z
Check the source to verify that this is the current version before use.

6.6.2 Electrical resistance.-This test shall be performed 100 percent. The test shall be performed as specified in 4.3.2, using test equipment specified in 4.2.4. Any primer which fails to comply with the requirement specified in 3.4 shall be classed defective and removed from the lot.

6.6.3 Functioning (Percussion-Electric Igniter, M30).- Sampling and disposition of the lot shall be as indicated in Standard MIL-STD-105, using Code letter L and an AQL of 0.25 percent. The M30 igniter shall be removed from the primer for testing. The test shall be performed as specified in Specification MIL-I-46574 (Ord).

6.6.4 Non-functioning (Percussion-Electric Igniter, M30).- Sampling and disposition of the lot shall be as indicated in Standard MIL-STD-105, using code letter J and an AQL of 0.65 percent. The test shall be performed as specified in Specification MIL-I-46574 (Ord). Igniters used in this test shall not be returned to the lot.

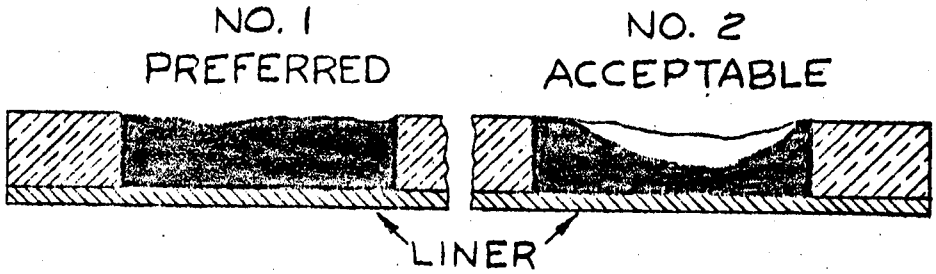
Custodian:
Army-MU

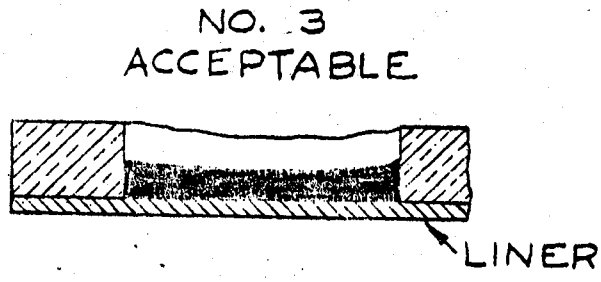
Preparing activity:
Army-MU

Project Number: 1390-A346

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:21Z
Check the source to verify that this is the current version before use.

ARTILLERY PRIMERS
STANDARDS FOR PAINT CONDITIONS IN FLASH HOLES





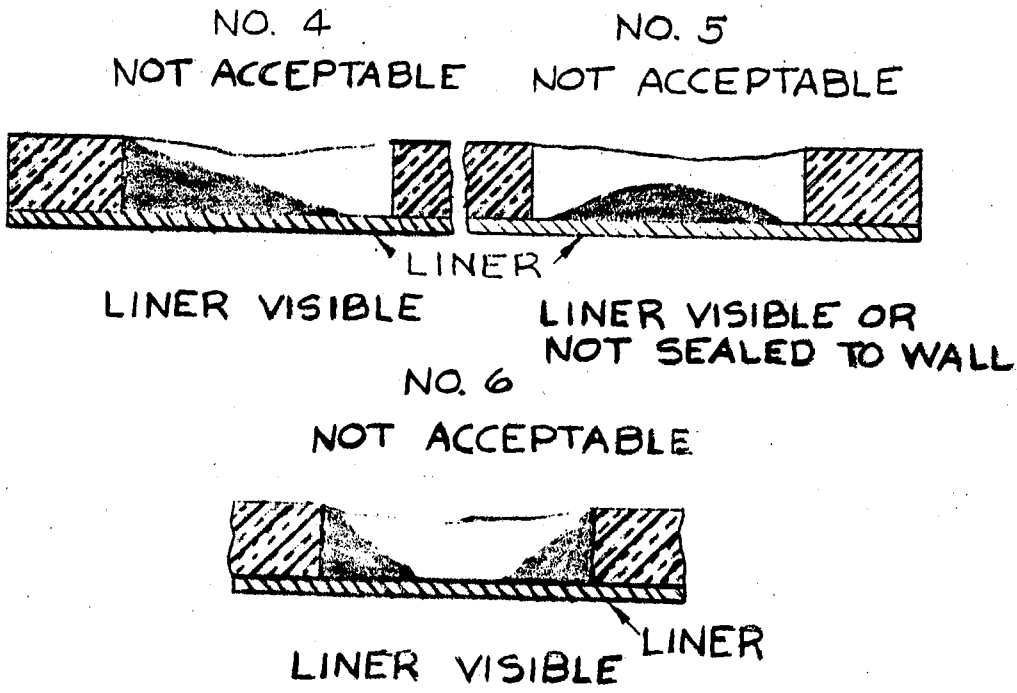


FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:21Z
Check the source to verify that this is the current version before use.